

Research Article

Brain Tumor Detection Using a Deep CNN Model

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The diagnosis of brain tumors through magnetic resonance imaging (MRI) has become highly significant in the field of medical science. Relying solely on MR imaging for the detection and categorization of brain tumors demands significant time, effort, and expertise from medical professionals. This underscores the need for an autonomous model for brain tumor diagnosis. Our study involves the application of a deep convolutional neural network (DCNN) to diagnose brain tumors from MR images. The application of these algorithms offers several benefits, including rapid brain tumor prediction, reduced errors, and enhanced precision. The proposed model is built upon the state-of-the-art CNN architecture VGG16, employing a data augmentation approach. The dataset utilized in this paper consists of 3000 brain MR images sourced from Kaggle, with 1500 images reported to contain tumors. Through training and testing, the pretrained CNN model achieves a precision and classification accuracy rate of 96%, and the loss is 1%. Moreover, it achieves an average precision, recall, and F1-score of 98.7%, 97.44%, and 98.06%, respectively. These evaluation metric values demonstrate the effectiveness of the proposed solution.

Keywords: brain tumor; CNN; MRI; VGG16

1. Introduction

Brain tumors stand as the 10th leading cause of mortality, impacting both adults and children. These tumors vary in type, texture, location within the brain, and shape, each presenting distinct challenges and often leading to lower survival rates. The complexity of brain tumor diagnosis and treatment is further underscored by factors such as tumor texture, location, and shape, which directly influence prognosis and treatment effectiveness. Therefore, achieving accurate and timely diagnosis, along with personalized treatment strategies, is paramount for enhancing outcomes in brain tumor patients. Precision in medical interventions becomes crucial, particularly when dealing with tumors that blur boundaries with healthy tissue. Accurately delineating tumor shape, boundary, and location is indispensable for ensuring safety during medical procedures, especially surgeries.

However, this task is inherently challenging due to the heterogeneous appearance of tumors and variations in

magnetic resonance imaging (MRI) image quality. This is where the transformative potential of artificial intelligence (AI) in medicine lies [1]. AI algorithms, notably those employing deep learning techniques such as convolutional neural networks (CNNs), have demonstrated remarkable proficiency in analyzing medical images, including MRI scans, often surpassing human expertise in accuracy and efficiency [2, 3]. Manual detection and segmentation, owing to their labor-intensive nature and susceptibility to subjective errors, are being supplanted by AI-driven approaches. There is a growing interest in leveraging supervised and unsupervised learning methods for brain tumor image detection and segmentation, offering promise for streamlining and improving these processes [4, 5].

Deep learning, particularly through automatic detection techniques, has emerged as a promising solution, showcasing impressive performance in various complex image tasks. CNNs, specifically, have become a cornerstone in medical image segmentation, leveraging their ability to learn

intricate features directly from images. This has established CNNs as leading algorithms in computer vision, especially in medical imaging applications. Their proficiency in detecting significant features within intricate images makes CNN-based approaches particularly well suited for tasks like brain tumor segmentation [6], consistently ranking highly in various image understanding challenges.

In this context, the proposed approach involves deploying a machine learning (ML) algorithm based on CNNs for brain tumor detection and classification [7–9]. By harnessing the capabilities of CNNs, this approach aims to elevate the accuracy and efficiency of brain tumor detection and classification, thereby contributing to enhanced medical diagnosis and treatment planning. Moreover, the development of robust brain tumor extraction methods based on image processing represents a significant advancement in medical imaging and healthcare. The primary contribution of this paper lies in devising a reliable method for detecting brain tumors from medical images using the VGG model and Adam optimizer. Such advancements hold promise for enabling early detection and treatment of brain tumors, ultimately leading to improved patient outcomes.

2. Related Works

In the medical field, the segmentation of brain tumors is a critical process that aids doctors in precisely identifying distinct parts of the tumor. To streamline this process, several automatic brain tumor detection techniques have been developed.

The authors in [3] proposed an efficient modified U-Net architecture, known as EMU-Net, which was created and used on the BraTS 2020 dataset. This dataset comprises a total of 369 patient images, although there have been reports indicating concerns regarding its quality. Indeed, segmentation plays a crucial role in medical imaging, especially in tasks like identifying and delineating brain tumors. Manual segmentation, while considered the gold standard, is labor-intensive and prone to variability between different observers.

In [10], the authors proposed a hierarchical deep learning-based brain tumor (HDL2BT) classification system that leverages CNNs for the detection and classification of brain tumors. This innovative system categorizes tumors into four distinct types: glioma, meningioma, pituitary, and no tumor. This suggested model demonstrates noteworthy performance metrics, achieving a precision of 92.13% and a miss rate of 7.87%.

The integration of deep learning and traditional ML techniques, as described in paper [11], involves a multistep process: feature extraction using deep learning techniques (AlexNet and ResNet-18), feature representation and classification using softmax, and SVM. By combining deep learning for feature extraction and traditional ML for classification, this approach aims to leverage the strengths of both methodologies to achieve robust and accurate classification results, particularly in complex tasks like medical image analysis, such as brain tumor segmentation and classification. However, if we assume that the accuracy

achieved is 95.10%, then it indicates a high level of performance in the classification task.

The study described in reference [12] tackled the difficult task of identifying brain tumors in MRI scans by leveraging a vast dataset of brain tumor images. The authors showcased the effectiveness of fine-tuning a cutting-edge YOLOv7 model via transfer learning, which led to substantial enhancements in detecting various types of brain tumors such as gliomas, meningiomas, and pituitary tumors. Deep learning models, such as CNNs, can effectively learn hierarchical representations of features directly from raw data, which is particularly beneficial when dealing with complex and high-dimensional data like medical images.

Reference [13] presents an efficient method for diagnosing brain tumors from MRI images utilizing deep convolutional neural networks (DCNNs). Brain tumors pose significant challenges in diagnosis and treatment, necessitating accurate and timely detection [7, 14]. The self-learning process involves the model automatically learning to discern relevant patterns and features during the training phase [15].

The primary limitation identified in the research conducted in [16] is their exclusive focus on binary classification of brain tumors, neglecting the consideration of multiclass classification [17, 18]. This limitation necessitates further analysis to ascertain the specific type of tumor, restricting the scope of the findings in terms of providing comprehensive information about different tumor categories.

Some widely recognized CNNs, including AlexNet [19], VGG [5], and ResNet [20], have gained prominence in the domain of medical image classification, particularly in tasks related to brain tumor classification. This section delves into recent studies that provide detailed insights into the application of these CNNs for the classification of brain tumors. In [1, 21–23], a DCNN architecture was employed for detecting the multiclass classification of brain tumors using MRI. In [24], the authors described the implementation of a DCNN for diagnosing brain tumors from MR images. The dataset used in this study comprises 253 brain MR images, with 155 images indicating the presence of tumors. The model developed in this research achieves an overall accuracy of 96% in correctly identifying MR images with tumors.

In [7], the authors introduced a novel deep feature fusion-based framework designed for classifying brain tumors from MR images. Data augmentation is crucial in training robust and effective deep learning models for detecting brain cancer from medical images such as MRI scans. This performance superiority over previous methods in detecting and augmenting brain tumors positions the proposed system as an impactful advancement in the field. The study aims to address several key motivations in the field of brain tumor diagnosis and classification. The authors in [8] investigated the efficacy of convolutional deep learning (CDL) methods and selected ML techniques for MRI-based brain tumor detection. Leveraging a dataset of MRI scans, preprocessing techniques such as skull stripping and intensity normalization were employed.

In reference [25], the authors utilized deep learning techniques to identify and segment brain tumor MRI

images. They employed a combination of CNN and fully convolutional network (FCN) methodologies in a sequential manner. The key discovery of their work is the utilization of YOLO-CNN for tumor area detection and localization, followed by segmentation of the identified tumor region using the FCN-U-Net architecture. This approach allowed for both accurate detection and precise delineation of the tumor boundaries in MRI images.

Several research papers have delved into leveraging adversarial approaches to tackle imbalanced data in ML [26, 27]. These methods are geared toward bolstering the robustness and generalization capabilities of deep learning models trained on imbalanced datasets by directly honing in on the decision boundary and training process. Specifically, in paper [26], the authors focus on confronting the challenge of imbalanced data classification through deep learning techniques enriched with noise augmentation.

Recent advancements include the use of EfficientNet and Vision Transformers (ViTs) for medical image classification tasks. EfficientNet, known for its scalable and efficient architecture, has been shown to provide high accuracy with fewer parameters compared to traditional CNNs. Studies such as [28] demonstrate the application of EfficientNet in brain tumor classification, achieving an accuracy rate of 96.5% on the BraTS 2019 dataset. ViTs, which rely on self-attention mechanisms rather than convolutional layers, have also been explored for their potential in medical imaging. Reference [29] discusses the use of ViTs in brain tumor classification, achieving competitive results with an accuracy rate of 97.2% on a similar dataset.

In Table 1, a comparative analysis of the performances of the related methods, including EfficientNet and ViTs, is provided.

These studies integrate noise into the training process of deep learning models to improve classification performance on imbalanced datasets. The introduction of noise encourages the model to acquire more robust and discriminative features, particularly for minority class samples, mitigating the effects of class imbalance.

Our model aims to accurately detect brain tumors and classify them using various techniques involving medical image processing, pattern analysis, and computer vision using the VGG model and Adam optimizer. By incorporating recent advancements such as EfficientNet and ViTs, our approach leverages the strengths of these state-of-the-art models to enhance the segmentation, classification, and overall diagnostic process for brain tumor identification.

3. Real-World Applications and Challenges

3.1. Performance Under Different MRI Machines. One significant aspect of deploying brain tumor classification models in real-world settings is their performance across MRI machines from different manufacturers and varying technical specifications. MRI machines differ in terms of magnetic field strength (1.5 T, 3 T, and 7 T), resolution, and imaging protocols, which can affect the quality and characteristics of the MRI images.

- **Magnetic field strength:** Higher magnetic field strengths generally provide better resolution and signal-to-noise ratio, which can enhance the model's performance. However, models trained on high-field strength images might not perform as well on images from lower-field strength machines without proper domain adaptation techniques.
- **Resolution and protocols:** Different resolutions and imaging protocols (e.g., T1-weighted, T2-weighted, and FLAIR) impact the appearance of brain tumors in the images. Models need to be robust enough to generalize across these variations or be retrained/fine-tuned on diverse datasets to ensure consistent performance.

To address these issues, techniques like transfer learning, domain adaptation, and data augmentation can be employed to adapt models to different imaging conditions. Ensuring a diverse training dataset that includes images from various MRI machines and settings can also help improve the model's generalizability.

3.2. Patient Demographics. Patient demographics, including age, gender, and genetic background, can influence the presentation and characteristics of brain tumors. For example, certain brain tumors are more prevalent in specific age groups or genders.

- **Age:** Pediatric brain tumors differ significantly from those in adults, both in terms of tumor types and imaging characteristics. Models trained on adult data might not perform well on pediatric cases without additional training.
- **Gender and genetic background:** Variations in genetic factors and gender-specific prevalence of certain tumor types can also affect model performance. It is crucial to include a balanced representation of demographic groups in the training dataset to ensure the model is unbiased and performs well across different patient populations.

3.3. Clinical Settings. The application of brain tumor classification models in clinical settings must consider the workflow and integration with existing medical systems. Real-world deployment requires the following:

- **Integration with Picture Archiving and Communication System (PACS):** The model should seamlessly integrate with PACS to allow radiologists to access and analyze the predictions within their routine workflow.
- **User interface:** A user-friendly interface is essential for clinicians to interact with the model, review the results, and make informed decisions. This includes clear visualizations of tumor segmentations and classification outputs.
- **Regulatory and ethical considerations:** Compliance with regulatory standards (e.g., FDA and CE) and addressing ethical concerns related to patient data

TABLE 1: Comparative study between the related works.

Study	Dataset	Method	Accuracy rate (%)
[3]	BraTS 2020 dataset	Modified U-Net architecture	90.81
[10]	BraTS 2015 dataset	Hierarchical deep learning-based brain tumor (HDL2BT)	92.13
[11]	BraTS 2018 dataset	AlexNet + SVM	95.10
[13]	The dataset used is MRI images of Vietnamese people, including 123 patients	DenseNet201	94.14
		ResNet152V2	93.31
		MobileNetV3	91.21
		VGG19	92.88
[20]	The Cancer Genome Atlas Low Grade Glioma Collection (TCGA-LGG)	ResNet50	94
[24]	Kaggle dataset (253 images)	DCNN (VGG16)	96
[28]	BraTS 2019 dataset	EfficientNet	96.5
[29]	BraTS 2020 dataset	Vision transformer (ViT)	97.2

privacy and informed consent are critical for clinical adoption.

3.4. Practical Challenges and Model Adaptations. Deploying brain tumor classification models in real-world settings presents several practical challenges, including the following:

- **Data quality and annotation:** High-quality, annotated datasets are essential for training accurate models. Inconsistent or inaccurate annotations can lead to poor model performance. Collaborative efforts between radiologists and data scientists are necessary to ensure the quality of the training data.
- **Handling class imbalance:** Brain tumor datasets often exhibit class imbalance, with certain tumor types being underrepresented. Techniques such as oversampling, synthetic data generation, and weighted loss functions can help address this issue.
- **Model interpretability:** Clinicians need to understand the model's decision-making process to trust its predictions. Techniques like Gradient-Weighted Class Activation Mapping (Grad-CAM) can provide visual explanations for the model's predictions, aiding in interpretability.
- **Continuous learning and adaptation:** As new data become available, the model should be periodically retrained and updated to maintain its performance. This requires a robust pipeline for data collection, annotation, and model training.

3.5. Model Adaptations. To enhance the practical applicability of brain tumor classification models, several adaptations can be considered:

- **Multimodal data integration:** Combining MRI images with other data sources, such as genetic information, clinical history, and pathology reports, can improve model accuracy and provide a more comprehensive diagnosis.
- **Edge computing:** Implementing models on edge devices within MRI machines can enable real-time processing and faster turnaround times for diagnosis.
- **Federated learning:** This approach allows training models across multiple institutions without sharing patient data, addressing privacy concerns and enabling the use of diverse datasets for improved generalization.

By addressing these real-world challenges and incorporating adaptations, brain tumor classification models can be effectively deployed in clinical settings, enhancing the diagnostic process and improving patient outcomes.

4. Scalability in Brain Tumor Classification Models

Scalability is a crucial aspect of deploying brain tumor classification models in real-world clinical settings, as it

ensures that models can handle larger datasets and varying image qualities without compromising performance. Addressing scalability involves both computational efficiency and the ability to generalize across diverse datasets.

4.1. Handling Larger Datasets. As the availability of medical imaging data grows, models must be capable of efficiently processing and learning from large volumes of data. This scalability can be achieved through several strategies:

1. **Distributed computing:** Leveraging distributed computing frameworks, such as Apache Spark or TensorFlow's distributed training capabilities, allows for the parallel processing of large datasets across multiple nodes, significantly reducing training time.
2. **Data parallelism and model parallelism:** Implementing data parallelism, where the same model is trained on different subsets of data concurrently, and model parallelism, where different parts of the model are trained on different devices, can enhance scalability.
3. **Efficient data management:** Utilizing optimized data storage solutions, such as hierarchical data formats (HDF5) or databases designed for large-scale image storage, ensures efficient data retrieval and preprocessing.
4. **Incremental learning:** Implementing incremental learning techniques allows the model to update itself with new data without retraining from scratch, making it feasible to handle continuously growing datasets.

4.2. Varying Image Qualities. Medical images from different sources often exhibit variations in quality due to differences in MRI machines, imaging protocols, and patient conditions. Ensuring that models can generalize across these variations is essential for robust performance:

1. **Domain adaptation:** Techniques such as domain adaptation and transfer learning can be employed to adapt models trained on one dataset to perform well on another dataset with different image characteristics. This involves fine-tuning the model on a small subset of the target domain data.
2. **Data augmentation:** Applying data augmentation techniques, such as rotation, scaling, and contrast adjustment, helps in simulating various image conditions during training, thus enhancing the model's ability to generalize across different image qualities.
3. **Multiresolution analysis:** Incorporating multi-resolution analysis, where the model processes images at different resolutions, can help in capturing both fine and coarse details, improving performance on images with varying resolutions.
4. **Preprocessing pipelines:** Developing robust preprocessing pipelines that include steps such as noise reduction, normalization, and alignment ensures that images fed into the model are of consistent quality, regardless of their source.

4.3. Cited Works on Scalability and Image Quality

1. Distributed deep learning for large-scale image datasets: Reference [30] discusses the implementation of distributed deep learning techniques to handle large-scale image datasets efficiently. The study highlights the importance of scalable computing resources and optimized algorithms to process vast amounts of data without sacrificing performance.
2. Handling diverse medical image datasets: Reference [31] explores methods for improving the generalization of medical image classification models across diverse datasets. The paper emphasizes domain adaptation and robust preprocessing techniques to manage variations in image quality.
3. Scalable CNNs for image classification: Reference [32] presents a scalable approach to CNNs that can efficiently process large image datasets. The research underscores the significance of model architecture optimization and efficient data handling in achieving scalability.
4. Transfer learning and data augmentation for medical imaging: Reference [33] discusses the application of transfer learning and data augmentation to enhance model performance on medical images with varying qualities. The study provides insights into how these techniques can improve the robustness and accuracy of classification models.
5. Efficient medical image analysis using distributed systems: Reference [34] focuses on the use of distributed systems for efficient medical image analysis. The paper highlights the role of scalable computing frameworks in managing large datasets and ensuring consistent model performance across different imaging conditions.

By implementing these strategies and leveraging insights from recent research, brain tumor classification models can achieve scalability, efficiently handling larger datasets and varying image qualities. This scalability is crucial for real-world applications, where the diversity and volume of medical imaging data continue to grow.

5. Integration Challenges in Brain Tumor Classification Models

Integrating brain tumor classification models into existing healthcare systems presents several challenges that must be addressed to ensure smooth adoption and effective utilization. Key issues include compatibility with current healthcare infrastructure, data interoperability, and workflow integration.

5.1. Compatibility With Current Healthcare Systems

5.1.1. Electronic Health Record (EHR) Systems. Integrating AI models with EHR systems is crucial for accessing patient history, storing results, and facilitating seamless workflows.

However, EHR systems vary widely in their design and functionality, leading to compatibility challenges.

- **Standardization:** Ensuring that the AI models can communicate with different EHR systems requires adherence to standardized protocols, such as Health Level 7 (HL7) and Fast Healthcare Interoperability Resources (FHIR).
- **API integration:** Developing robust APIs that allow AI models to interface with EHR systems enables real-time data exchange and integration. This includes the ability to fetch patient data, update records with AI-generated results, and trigger alerts for clinicians.

5.1.2. PACS. PACS is used to store and access medical imaging data, making its integration with AI models essential for processing and analyzing MRI images.

- **Digital Imaging and Communications in Medicine (DICOM) compatibility:** Ensuring the AI models can handle DICOM standards is critical for seamless image retrieval and storage.
- **Workflow integration:** AI models should be integrated into the PACS workflow to enable automatic processing of images as they are acquired, providing real-time analysis and feedback to radiologists.

5.2. Data Interoperability. Data interoperability is the ability to exchange and use information across different systems and platforms. It is a significant challenge in healthcare due to the diversity of data formats, standards, and sources.

5.2.1. Heterogeneous Data Sources. Medical data are collected from various sources, including different imaging modalities, laboratory tests, and clinical notes. Ensuring interoperability across these heterogeneous sources is essential for comprehensive analysis.

- **Unified data formats:** Converting data from different formats into a unified, standard format helps in seamless integration and analysis. This often involves preprocessing steps such as normalization, standardization, and annotation harmonization.
- **Metadata handling:** Proper handling of metadata, including patient demographics, imaging parameters, and clinical context, is crucial for accurate analysis and interpretation of results.

5.2.2. Semantic Interoperability. Beyond syntactic interoperability, which deals with data formats, semantic interoperability ensures that the meaning of the data is preserved and understood consistently across systems.

- **Ontologies and standards:** Utilizing medical ontologies and standards like Systematized Nomenclature of Medicine—Clinical Terms (SNOMED CT) and Logical Observation Identifiers Names and Codes (LOINC) helps in achieving semantic interoperability.

- Mapping and transformation: Implementing mapping and transformation tools that convert data between different standards and ontologies ensures that the AI models can understand and use the data correctly.

5.3. Workflow Integration. Successfully integrating AI models into clinical workflows requires careful consideration of how these tools will be used by healthcare professionals.

5.3.1. User Interface and Experience. Developing intuitive and user-friendly interfaces for AI tools is critical for their adoption by clinicians. This includes clear visualizations, easy navigation, and actionable insights.

- Visualization tools: Providing advanced visualization tools that highlight AI-detected anomalies, segmentations, and classification results helps clinicians quickly interpret the findings.
- Interactive interfaces: Allowing clinicians to interact with the AI model, such as adjusting parameters or verifying results, enhances trust and usability.

5.3.2. Clinical Decision Support. AI models should complement, not replace, the clinician's expertise. Integrating decision support features that provide recommendations while allowing clinicians to make the final decisions is key.

- Alert systems: Implementing alert systems that notify clinicians of critical findings or discrepancies ensures timely intervention.
- Explainability: Ensuring that AI models provide explanations for their predictions helps clinicians understand and trust the results. Techniques like Grad-CAM and Local Interpretable Model-Agnostic Explanations (LIME) can be used to provide visual and textual explanations.

5.3.3. Regulatory and Ethical Considerations. Compliance with regulatory standards (e.g., FDA and CE) and addressing ethical concerns related to patient data privacy and informed consent are critical for clinical adoption.

- Data privacy: Implementing robust data privacy measures, such as encryption and anonymization, ensures that patient information is protected.
- Regulatory approval: Obtaining regulatory approval for AI models involves rigorous testing and validation to ensure their safety and efficacy in clinical settings.

By addressing these integration challenges, brain tumor classification models can be effectively deployed in healthcare systems, enhancing diagnostic accuracy and improving patient outcomes.

6. Dataset Description

The dataset used in this study is Brain MRI Images for Brain Tumor Detection obtained from <https://kaggle.com> [6]. It

comprises three folders: The first folder contains 1500 MRI scans showcasing brain tumors and the second folder contains 1500 MRI scans of healthy brains. Additionally, there is a folder containing unlabeled MRI scans intended for testing purposes. Consequently, the final database is constructed from the first two folders, comprising 3000 images in total. These images are distributed as follows: 1500 images with tumors and 1500 images without tumors. Figure 1 illustrates sample images from the dataset. It contains two MRI scan images:

- MRI scan with brain tumor: This image shows a brain MRI scan with a visible tumor. The highlighted area indicates abnormal tissue growth, which the model needs to detect and classify accurately.
- MRI scan of a healthy brain: This image depicts a brain MRI scan without any signs of a tumor. It represents normal brain tissue, helping the model to differentiate between healthy and tumor-affected regions.

6.1. Data Augmentation and Image Preprocessing. To improve contrast intensity, normalization techniques are employed to narrow down intensity values to a stable range, thereby reducing variability. This process aims to standardize the mean intensity value to zero and the standard deviation toward one. Initially, the images undergo thresholding using a threshold value of 45, followed by a sequence of erosions and dilations to eliminate small regions of noise. Subsequently, the images are normalized by identifying the largest contour within each image and cropping the images based on the extreme points of the contour in the top, bottom, left, and right directions. This normalization process helps to enhance the quality and consistency of the images, facilitating more accurate analysis and interpretation. The entire normalization process is illustrated in Figure 2 that contains 4 captions:

- Caption 1: original image—This is the initial brain MRI scan before any processing. This image contains variable intensity values and noise that can affect subsequent analysis.
- Caption 2: finding the biggest contour—The largest contour within the image is identified. This contour highlights the main region of interest in the scan, typically the brain or significant features.
- Caption 3: identifying extreme points—The extreme points (top, bottom, left, and right) of the largest contour are determined. These points define the boundaries for cropping, focusing on the region of interest.
- Caption 4: cropped image—The MRI scan is cropped based on the extreme points of the largest contour. This step enhances the image by removing irrelevant background and standardizing the region of interest, preparing it for accurate analysis.

These captions explain each step in the normalization process, demonstrating how it improves image quality and consistency for more reliable analysis and interpretation.

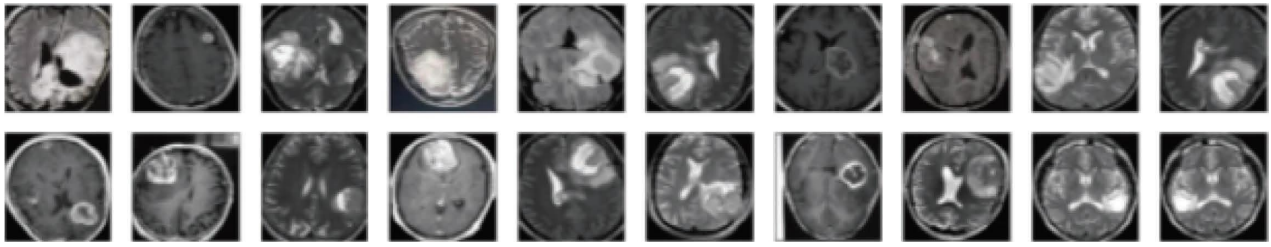


FIGURE 1: Samples of the dataset.

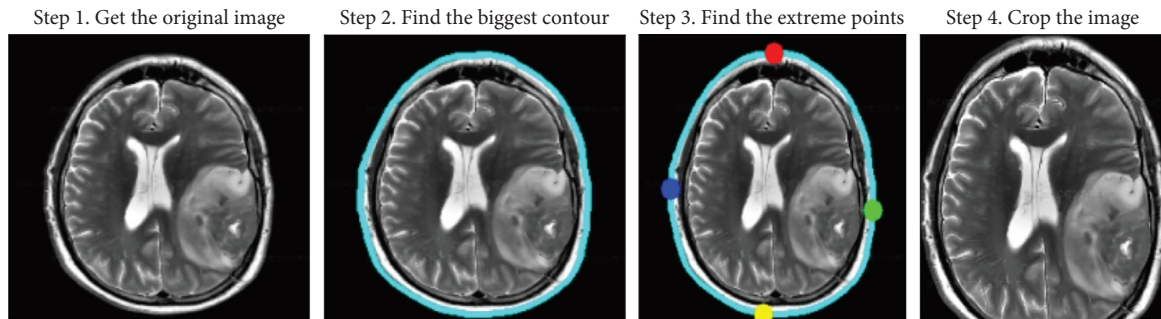


FIGURE 2: Brain MRI images in a normalization process.

Data augmentation methods [35] prove beneficial in addressing the imbalance and scarcity of data in certain classes when dealing with limited and uneven datasets. This approach proves particularly useful for achieving a balance in the number of images across different MRI classes related to brain tumors and for expanding the overall dataset. Various augmentation techniques, including rotation, cropping, height and width adjustments, filling operations, zooming, and horizontal rotation brightening, are employed to augment images and rectify class imbalances. The MR images obtained from the dataset exhibited varying sizes. As these images served as the input layer for the networks, they were resized to dimensions of 80×80 pixels. To augment the dataset, each image underwent two transformations. Firstly, the images were rotated by 90° , and secondly, they were flipped vertically.

By applying these transformations, the chosen dataset was augmented threefold, resulting in a total of 9792 images for further analysis and processing.

Figure 3 provides a visual representation of a subset of image samples from the brain tumor dataset after the application of data augmentation. It contains five captions for five different images:

- Caption 1: original image—This is the initial brain MRI scan before any augmentation. It serves as the reference image for subsequent transformations.
- Caption 2: rotated image (90°)—The original MRI scan is rotated by 90° . This transformation helps in increasing the variability of the dataset, allowing the model to recognize brain tumors from different angles.
- Caption 3: vertically flipped image—The original MRI scan is flipped vertically. This augmentation technique

adds diversity to the dataset, improving the model's ability to generalize across different orientations.

- Caption 4: cropped image—This is the original MRI scan after cropping. Cropping focuses on specific regions of the image, which can help the model learn important features more effectively.
- Caption 5: zoomed image—This is the original MRI scan after applying zooming. Zooming in on the image allows the model to detect fine details within the scan, enhancing its accuracy in identifying tumors.

These captions describe each augmentation technique applied to the MRI images, illustrating how these transformations contribute to a more diverse and balanced dataset, essential for robust model training and accurate brain tumor detection.

7. Proposed Methodology

The proposed methodology utilizes a DCNN architecture as the foundation for brain tumor detection in brain MRI images. The approach comprises several key steps:

- Input and preprocessing: Initially, the brain MRI image serves as the input. The first step in preprocessing involves data normalization to ensure consistency across the images. This includes image thresholding to remove background noise and dilations to eliminate small noise regions. After noise reduction, images are further processed through normalization techniques to standardize the intensity values, bringing the mean intensity value to zero and the standard deviation to one. This step is crucial for stabilizing the training process and ensuring that the input data are in a consistent format.

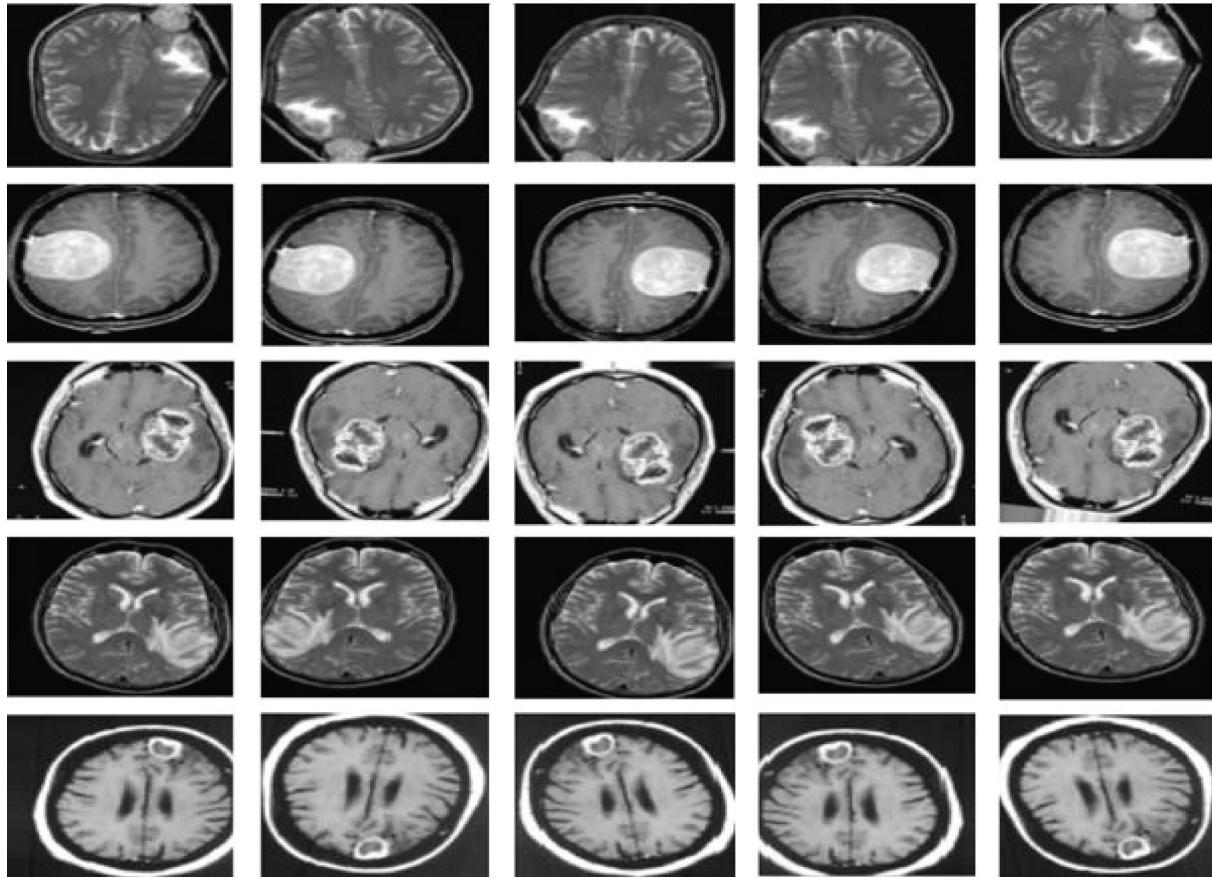


FIGURE 3: A set of image samples after applying data augmentation.

- **Data augmentation:** To address the issue of limited and imbalanced data, data augmentation techniques are employed. These techniques include rotation, cropping, height and width adjustments, filling operations, zooming, and horizontal rotation. The augmented dataset helps in improving the model's generalization by providing diverse variations of the input images, thereby reducing the risk of overfitting. Each image is resized to 80×80 pixels, and additional transformations, such as a 90° rotation and vertical flipping, are applied to further augment the dataset, tripling its size to 9792 images.
- **Feature extraction with VGG16:** Our study employs the VGG16 model as the feature extraction component. VGG16 is a renowned CNN-based model pretrained on extensive datasets like ImageNet, which contains over one million images. This pre-training equips VGG16 with the capability to effectively learn and extract relevant features from new images. The architecture of VGG16, illustrated in Figure 4, was originally proposed by [24], members of the Visual Geometry Group, during the 2014 ImageNet Competition. Their submissions secured first and second places in the fields of classification and localization, respectively. The use of VGG16 allows our model to leverage these prelearned features, enhancing its performance in classifying brain

MRI images. The architecture of VGG16 is illustrated in Figure 4.

- **Overfitting mitigation:** Overfitting correction is addressed through four techniques: data augmentation, dropout, batch normalization, and pooling. Data augmentation, as mentioned earlier, increases the diversity of the training data. Dropout involves randomly setting a fraction of input units to zero during training to prevent overreliance on certain features. Batch normalization standardizes the inputs to each layer, which accelerates training and improves model stability. Pooling layers reduce the dimensionality of the feature maps, retaining essential information while reducing computational complexity.
- **Model training with Adam optimizer:** The training process involves using the Adam optimization function, which is an adaptive learning rate optimization algorithm designed specifically for training deep neural networks. Adam combines the advantages of two other extensions of stochastic gradient descent, namely, Adaptive Gradient Algorithm (AdaGrad) and Root Mean Square Propagation (RMSProp). It computes adaptive learning rates for each parameter, which enhances training efficiency and convergence. The CNN model, incorporating VGG16 for feature extraction and the Adam optimizer, is trained to classify images into

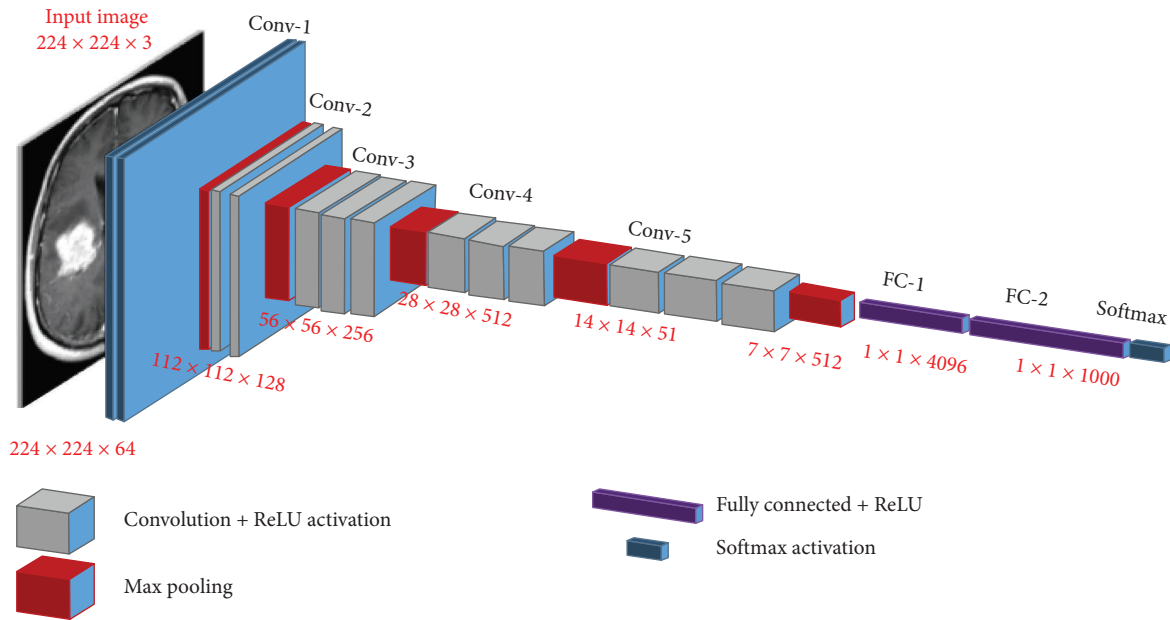


FIGURE 4: Architecture of VGG16 model [36].

two categories: YES (tumor present) and NO (no tumor).

- **Evaluation:** The performance of the trained model is evaluated using various metrics, including precision, recall, sensitivity, specificity, accuracy, and $F1$ -score. These metrics provide a comprehensive assessment of the model's effectiveness in detecting brain tumors from MRI images. The evaluation process involves testing the model on a separate dataset to ensure its generalization capability and robustness.

The proposed methodology, depicted in Figure 5, outlines the entire process from data collection and pre-processing to model training and evaluation. By integrating advanced techniques such as VGG16 for feature extraction, data augmentation to enhance the dataset, and the Adam optimizer for efficient training, our approach aims to achieve high accuracy in brain tumor detection while maintaining computational efficiency and robustness.

8. Results and Discussion

In this section, the experimental results of the proposed model will be presented. The process begins by uploading the MRI brain image. Upon submission of the MRI brain image, the model commences the analysis using learning models to ascertain the presence of a brain tumor. Upon successful identification of a tumor, a message will affirmatively declare "Brain Tumor Detected."

Figure 6 displays the training and validation accuracy of the proposed method using 32 epochs. Conversely, Figure 7 illustrates the loss function.

Our proposed model outperforms the previously developed approaches demonstrating accuracy of 98%. Based on the results depicted in the confusion matrix in Figure 8, it is observed that out of 600 cases, the model correctly

classified 305 positive cases of brain tumors and accurately identified 283 nontumorous brain images as belonging to the negative class. These values, namely, true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), obtained from the confusion matrix, are subsequently utilized by the code for calculating the loss metrics, including accuracy, precision, and sensitivity.

Our experimental results illustrate that the proposed CNN model exhibits superior convergence compared to the ANN approach, random forest classifier, transfer learning algorithms, and other CNN models. As indicated in Table 2, our model attained the highest accuracy rate of 98% and the best $F1$ -score of 98.06%, along with a precision of 98.7%.

The values attained by our proposed CNN model stand out for their exceptional performance, surpassing those achieved by the previously mentioned models. Through a comparative analysis presented in Table 3, it becomes evident that our model outperforms others, securing the highest rank in terms of accuracy. This underscores the efficacy and reliability of our approach in accurately detecting brain tumors from MRI scans.

While our proposed CNN model demonstrates outstanding performance with a high accuracy rate of 98%, there are several limitations and potential risks that should be considered:

- **Computational overhead:** The CNN model, especially when using advanced architectures and optimizers like the Adam optimizer, can be computationally intensive. Training such models requires significant computational resources, including high-performance GPUs and extensive memory. This can pose a challenge for institutions or individuals with limited access to these resources.
- **Trade-offs between accuracy and efficiency:** Achieving high accuracy often involves complex models with

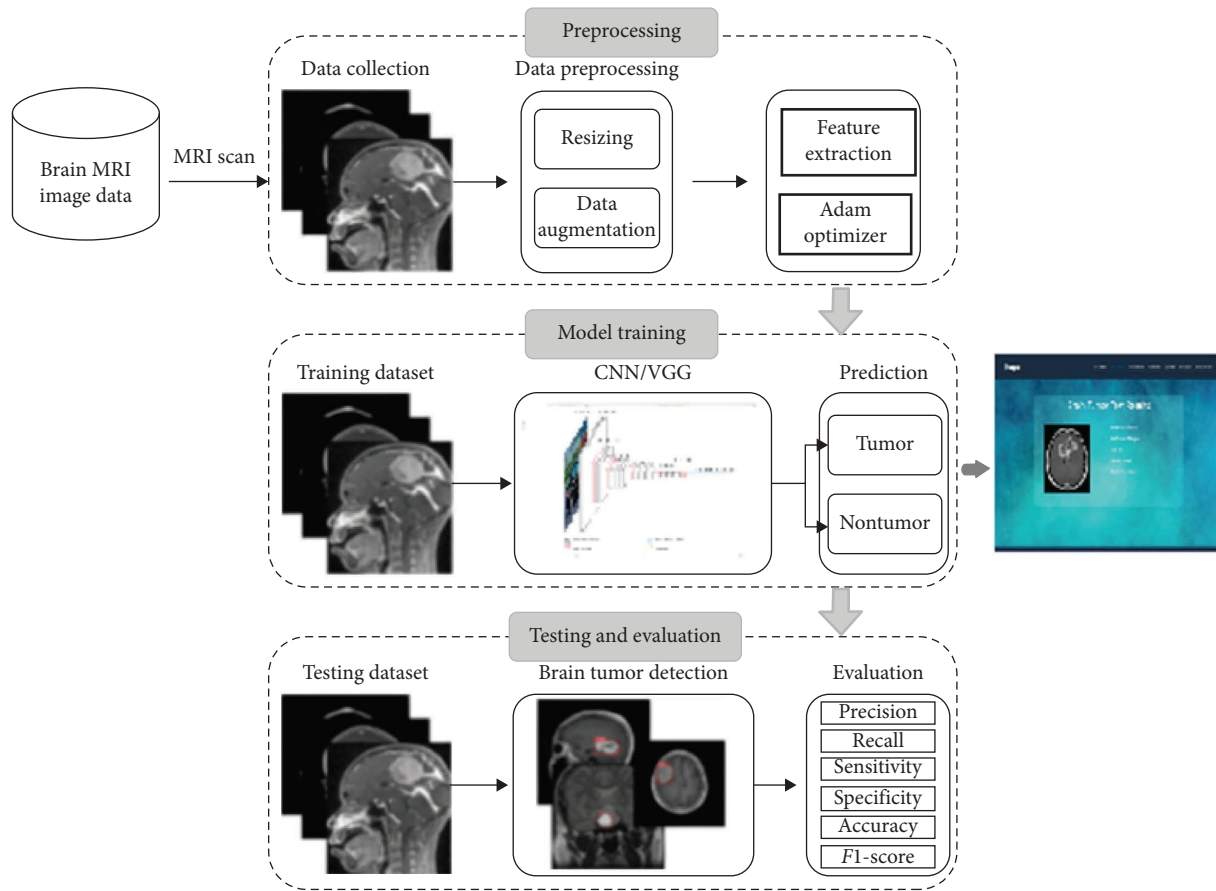


FIGURE 5: Proposed model methodology.

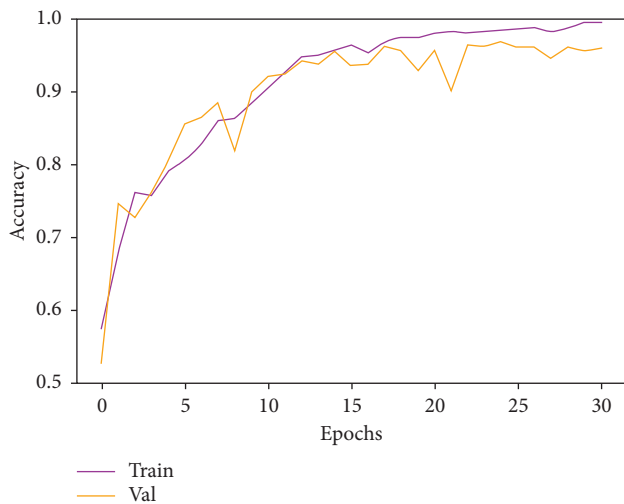


FIGURE 6: The training and validation accuracy over the epochs.

a large number of parameters, leading to increased training and inference times. While our model achieves superior accuracy, the trade-off is that it may be less efficient in terms of speed and computational cost compared to simpler models. For real-time applications or deployment in resource-constrained environments, this efficiency trade-off

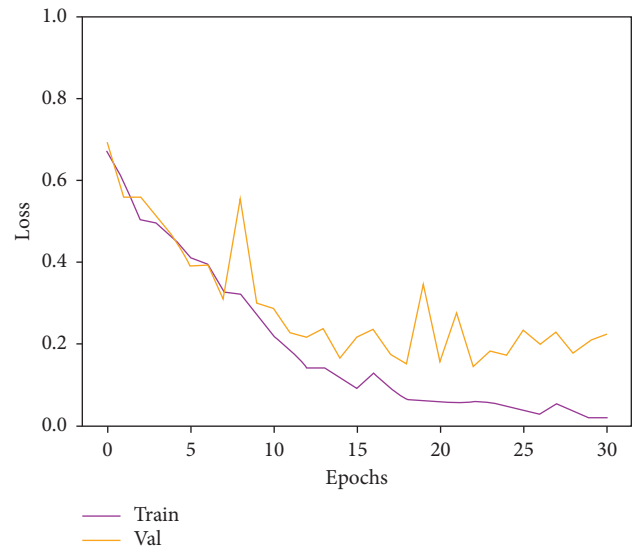


FIGURE 7: The training and validation loss over the epochs.

must be carefully balanced against the need for high accuracy.

- **Data dependency:** The performance of our model heavily depends on the quality and quantity of the training data. While data augmentation techniques

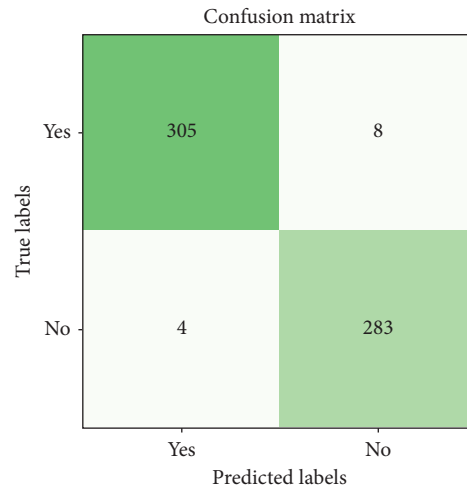


FIGURE 8: Confusion matrix.

TABLE 2: Performance of the proposed model.

Evaluating metrics	Performance (%)
Precision	98.7
Sensitivity	97.44
F1-score	98.06
Specificity	97.25

have been employed to enhance the dataset, any biases or inconsistencies in the data can affect the model's generalization ability. Additionally, the model may not perform as well on data from different sources or with different imaging protocols.

- **Risk of overfitting:** Although our model has shown excellent results on the validation set, there is always a risk of overfitting, especially with high-capacity models. Overfitting occurs when the model learns the noise in the training data instead of the actual underlying patterns, which can lead to poor performance on unseen data. Regularization techniques and cross-validation are essential to mitigate this risk.
- **Interpretability:** Deep learning models, including CNNs, are often seen as "black boxes" due to their complex nature. Understanding and interpreting the decision-making process of the model can be challenging, which is a significant limitation in critical applications like medical diagnosis. Efforts to improve the interpretability of the model's predictions are necessary to build trust and facilitate clinical adoption.

Addressing computational overhead and trade-offs involves optimizing model efficiency, leveraging specialized hardware, employing hybrid approaches, and implementing scalable training techniques:

- **Optimizing model architecture:** To mitigate computational overhead, optimizing the model architecture for efficiency without compromising accuracy is crucial. Techniques such as model pruning,

quantization, and knowledge distillation can be employed to reduce the model size and enhance inference speed.

- **Utilizing efficient hardware:** Leveraging specialized hardware like Tensor Processing Units (TPUs) or optimizing the model for deployment on edge devices can help balance the trade-off between accuracy and efficiency.
- **Hybrid approaches:** Combining our CNN model with other ML techniques, such as assembling with less computationally intensive models, can achieve a balance between high accuracy and efficiency.
- **Scalable training:** Implementing scalable training techniques, such as distributed training and leveraging cloud-based platforms, can address the computational demands during the training phase.

Ensuring ethical standards in our research is paramount, especially when dealing with sensitive medical data. We have implemented stringent measures to comply with ethical guidelines and regulations. All patient data used in this study were obtained with explicit informed consent, guaranteeing that participants were fully informed about the study's objectives, procedures, and the use of their data. This process involved providing clear and detailed information to participants, ensuring they understood their rights and the purpose of the research:

- **Data handling and patient consent:** Handling medical data involves significant ethical considerations. Ensuring patient consent for the use of their data in research is paramount. Our study strictly adheres to ethical guidelines by obtaining informed consent from all participants whose data were included in the dataset. Data anonymization techniques are employed to protect patient privacy and confidentiality.
- **Compliance with ethical standards:** Our research complies with all relevant ethical standards and regulations, including institutional review board (IRB) approvals where applicable. Ensuring the ethical use of

TABLE 3: Comparative study.

Study	Method	Accuracy rate (%)
[3]	Modified U-Net architecture	90.81
[10]	Hierarchical deep learning-based brain tumor (HDL2BT)	92.13
[11]	AlexNet + SVM	95.10
[13]	DenseNet201	94.14
	ResNet152V2	93.31
	MobileNetV3	91.21
	VGG19	92.88
[20]	ResNet50	94
[24]	DCNN (VGG16)	96
Proposed model	CNN + Adam optimizer	98

data and maintaining transparency in our research practices is fundamental to the integrity of our study.

Although our proposed CNN model achieves state-of-the-art performance in brain tumor detection, acknowledging and addressing its limitations and potential risks is essential for practical deployment. By optimizing computational efficiency, ensuring high-quality data, mitigating overfitting, and enhancing interpretability, we can improve the model's robustness and applicability in real-world scenarios. Additionally, addressing ethical implications through proper data handling and patient consent ensures compliance with ethical standards and maintains the integrity of our research.

9. Conclusion

This paper introduces a CNN model tailored for segmenting MRI images depicting brain tumors into two classes: those with tumors and those without. Its primary contribution lies in the development of a robust method for detecting brain tumors from medical images, potentially leading to significant advancements in early tumor detection and treatment, thus improving patient outcomes.

The proposed model is built upon the state-of-the-art CNN architecture VGG16, employing a data augmentation approach. Our primary objective is to utilize CNNs as a ML approach for brain tumor detection and classification. Furthermore, we advocate for the integration of the Adam optimization function to construct predictive models utilizing a dataset comprising 3000 high-resolution MRI images, with subsequent evaluation employing various metrics. Results demonstrate the superior accuracy of the proposed model compared to other neural network models, achieving an impressive accuracy rate of 98%. CNN emerges as the most effective technique for predicting brain tumors within the dataset. The study includes a detailed analysis of performance evaluation metrics and curve analysis, showcasing the CNN network's proficiency in detecting and classifying brain tumors with heightened accuracy. The paper elucidates the architecture of CNNs and highlights their efficacy when applied to a refined database of brain images.

Future research should focus on the integration of IoT and AI to significantly advance healthcare applications, particularly in the realm of brain tumor detection and

management. The deployment of IoT-enabled wearable devices and smart sensors for continuous monitoring can revolutionize how patient data are collected and analyzed. These devices can track vital signs and neurological activity in real time, providing a continuous stream of data that AI algorithms can process to detect early signs of brain tumors or neurological changes. Such real-time monitoring systems can facilitate early diagnosis and intervention, significantly improving patient outcomes. Additionally, the development of AI-driven predictive models using these data can help in identifying at-risk patients and enabling preventive measures tailored to individual needs.

Moreover, integrating AI with IoT in diagnostic imaging devices can enhance the accuracy and efficiency of brain tumor diagnosis. Smart imaging devices equipped with edge AI capabilities can perform real-time image processing and analysis, reducing the time between image acquisition and diagnosis. This can be particularly beneficial in emergency and remote settings where quick and accurate diagnostics are critical. Future research can also explore the creation of unified health data platforms that integrate data from various IoT devices, EHRs, and imaging systems. Such platforms would ensure seamless data exchange and create comprehensive patient profiles, aiding in more informed and holistic clinical decision making. Addressing ethical considerations, such as data privacy, informed consent, and bias in AI models, will be essential in building patient trust and ensuring compliance with regulatory standards. By focusing on these areas, future research can pave the way for more effective, personalized, and equitable healthcare solutions.

Appendix

The code was collected from Kaggle, available at this link: <https://www.kaggle.com/code/arunchechi/brain-tumor-detection-using-cnn>.

Data Availability Statement

The datasets were collected from Kaggle, available at this link: <https://www.kaggle.com/datasets/abhranta/brain-tumor-detection-mri>.

Conflicts of Interest

The authors declare no conflicts of interest.

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